

JEE MAIN REVISION MASTER: SOLUTIONS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 188 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Solutions - Core Application	188 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Solutions - Core Application (Very High Yield — 188 PYQs)

Core Principle: Key problem pattern in Solutions.

■ ■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Given below are two statements : Given : Molar mass of C,H,O,Cl are 12,1,16 and 35.5gmol⁻¹ , respectively Statement I : In 30%(w/w..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "At 27°C,0.1M,1LK_(4)[Fe(CN)_(6)] aqueous solution and 0.1M,1LFeCl_(3) aqueous solution are placed in a container separated by a s..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "At T(K),2 moles of liquid A and 3 moles of liquid B are mixed. The vapour pressure of ideal solution formed is 320 mm Hg . At this stage, ..." [Ans: C]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 28th January Morning Shift (Q#1)

At T(K), 2 moles of liquid A and 3 moles of liquid B are mixed. The vapour pressure of ideal solution formed is 320 mm Hg. At this stage, one mole of A and one mole of B are added to the solution. The vapour pressure is now measured as 328.6 mm Hg. The vapour pressure (in mm Hg) of A and B are respectively:

- (A) 400, 300
- (B) 600, 400
- (C) 500, 200 **[Correct]**
- (D) 300, 200

Correct Answer: C

Solution Analysis: For an ideal solution, by Raoult's law: $P_{\text{total}} = x_{\text{A}}P_{\text{A}}^0 + x_{\text{B}}P_{\text{B}}^0$ where $P_{\text{A}}^0, P_{\text{B}}^0$ are vapour pressures of pure liquids A and B. Step 1: First mixture (2 mol A + 3 mol B) Total moles = 2+3=5 $x_{\text{A}} = (2/5), x_{\text{B}} = (3/5)$ Given $P_{\text{total}} = 320$ mm Hg: $320 = (2/5)P_{\text{A}}^0 + (3/5)P_{\text{B}}^0$ Multiply by 5: $1600 = 2P_{\text{A}}^0 + 3P_{\text{B}}^0 \dots (1)$ Step 2: After adding 1 mol A and 1 mol B New moles: A = 3, B = 4, total = 7 $x_{\text{A}} = (3/7), x_{\text{B}} = (4/7)$ Given new vapour pressure = 328.6 mm Hg: $328.6 = \frac{3}{7}P_{\text{A}}^0 + \frac{4}{7}P_{\text{B}}^0$